```
local markings = {}

function unique (marking)
  for _, m in ipairs (markings) do
    if marking == m then
      return m
    end
  end
  markings [#markings+1] = marking
  return marking
end
```

```
function reachable ()
  local explored    = {}
  local encountered = {
    [unique (initial)] = true
  }
  while next (encountered) do
    local marking = next (encountered)
    encountered [marking] = nil
    if not explored [marking] then
      explored [marking] = true
      for _, transition in ipairs (transitions) do
        local successor = transition (marking)
        if successor then
          successor = unique (successor)
          marking     [transition] = successor
          encountered [successor ] = true
        end
      end
     end
  end
  return explored
end
```